# Complexity_Simplified_N.txt - Step-by-Step Guide

For: Mitchell Ray - secp256k1 Curve Calculations

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## 1. PROBLEM IDENTIFICATION

You are using the WRONG exponent for cube roots modulo N.
You computed: omega2 = RQ^((N-1)/3) mod N
This gives a PRIMITIVE CUBE ROOT OF UNITY, not a cube root of Lambda^3.

## 2. THE CORRECT METHOD

For cube roots mod N: x = a^((2*N-1)/3) mod N
Not (N-1)/3 which gives omega

## 3. YOUR CORRECT VALUES

N = 115792089237316195423570985008687907852837564279074904382605163141518161494337
Lambda = 97451685862885086182458552040892158509924235661624603229050850812487253689501
p1 = 1248780847746852317428964695904392891045016528862400526454142780194939123483
p2 = 21551977082208859489759061364299864038123955443494189974630776168682352336746
p3 = 92991331307360483616382958948483650923668592306718313881520244192640870034108

## 4. YOUR MISSING CALCULATIONS

Cube root of Lambda^3 mod N = Lambda
Lambda^3 mod N = 38932995473618115921409207338423707925309087193404485552072959838229500524277

Cube roots of Lambda^3 * (p-N) mod N:
Root 1 = Lambda * p1 mod N = 21864327243976606572352071181633865121353765839471190989110739184573495121925
Root 2 = Lambda * p2 mod N = 90798428535687044113399170261082130318391557661176307034657430351294071817198
Root 3 = Lambda * p3 mod N = 3129333457652544737819743565971912413092240778427406358836993605650594555214

## 5. THE ERROR WITH y1, y2, y3

y1^3, y2^3, y3^3 all labeled as omega^2-1 but are DIFFERENT values
Correction: Use exponent (2*N-1)/3 for cube roots, not (N-1)/3

## 6. NEXT STEPS

1. Use exponent (2*N-1)/3 for cube roots of values
2. Add the 3 cube roots of Lambda^3 * (p-N) to your file
3. Fix y1, y2, y3 calculations
4. Verify all calculations

## 7. KEY FORMULA

Cube root of a mod N = a^((2*N-1)/3) mod N

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